Best Model Architecture & Comparative Benchmark Dossier

Pedagogical Walkthrough of FiLM-RGCN, Domain Knowledge Weighting, and 10-Model Benchmark

Executive Summary: All 10 models evaluated in this research were implemented, trained, and benchmarked by our team under an identical 5-fold cross-validation protocol across the 364 curated datapoints. This document explains the inner workings of our top-performing model---the **pH-Conditioned Multi-Relational Graph Neural Network with Dual-Scale Distance Readout (FiLM-RGCN)** (**46.35 mV MAE, R^2 = 0.929**)---explaining step-by-step how domain knowledge weights were assigned to high-importance features and comparing it against all other model iterations.

1. Step-by-Step Walkthrough of the Best Architecture (FiLM-RGCN)

Step 1: Invariant Node Vector & Domain Knowledge Weighting (49 Dimensions):
In our preliminary XGBoost and physical analyses, we discovered that certain features play an overwhelming role in redox tuning. Instead of treating all properties as equally flat numbers, we assigned **domain knowledge scaling factors** to emphasize critical physical channels:

- **Flexibility (Scaled 5.0×):** In our XGBoost feature gain analysis, backbone flexibility accounted for **49.2% of total model gain**. In Marcus electron transfer theory, local flexibility dictates the reorganization energy (λ), determining how easily the active site accommodates charge changes. Multiplying flexibility by 5.0 gives the neural network an immediate inductive prior.
- **CYS, TRP, PRO (Scaled 3.0×):** Cysteines provide direct covalent thiolate coordination; Tryptophans provide bulky hydrophobic shielding; Prolines enforce rigid active-site loop geometry. These 3 residues were scaled 3.0×.
- **Coordinating CYS Flag (Scaled 5.0):** Directly ligating cysteines ($S_\gamma \leq 3.0$ Å to Fe) receive an explicit 5.0 indicator.
- **Other 27 Physicochemical Properties (1.0×):** Hydrophobicity, isoelectric point, volume, SASA, and secondary structure propensities min-max normalized to [0, 1].

Step 2: The 5 Biophysical Edge Types (Multi-Relational Graph):
Rather than treating the protein as a generic unweighted contact map, message passing is routed through 5 explicit physical relation channels: **R0 (Covalent Backbone)**, **R1 (Spatial Contacts, d ≤ 8.0 Å)**, **R2 (Hydrogen Bonds, $d_{NO} \leq 3.5$ Å)**, **R3 (Disulfide Bridges, $d_{SS} \leq 2.5$ Å)**, and **R4 (Ionic Salt Bridges, d ≤ 6.0 Å)**. Each relation has its own dedicated parameter weight matrix W_r.

Step 3: Relational Graph Convolutions (RGCN Message Passing):
For each layer $l \in \{1, 2\}$ (hidden dimension $h = 7$), the residue embeddings are updated by aggregating messages across all 5 relation types:

$$z_i^{(l)} = W_0\, h_i^{(l-1)} + \sum_{r=0..4} (1/|N_i^r|) \sum_{j \in N_i^r} W_r\, h_j^{(l-1)}$$

Followed by 1D Batch Normalization and ReLU activation.

Step 4: Layer-Wise pH FiLM Generator (Environmental Thermostat):
Solution acidity (pH) physically alters amino acid ionization states and electrostatic field strength. Instead of simply concatenating pH at the output, our **Feature-wise Linear Modulation (FiLM)** generator dynamically alters residue embeddings after every RGCN layer:

$$[\gamma, \beta] = W_2\, \mathrm{ReLU}(W_1\, pH_{norm} + b_1) + b_2$$
$$h_i^{(l)} = \mathrm{Dropout}(\mathrm{ReLU}((1.0 + \gamma) \blacksquare h_i^{(l)'} + \beta), p=0.1)$$

Zero-initializing W_2 ensures clean initial identity propagation while learning non-linear pH-dependent ionization shifts.

Step 5: Dual-Scale Distance-Exponential Soft Readout (Active-Site Attention):
Standard graph pooling (global mean pool) averages all 300+ residues equally, washing out the active site. Our model computes Euclidean distance d_i^{FeS} from every residue to the FeS cluster and applies two learnable exponential soft decay operators:

$$w_{i,\,tight} = exp(-\gamma_{tight} \cdot d_i^{FeS}) / \sum_j exp(-\gamma_{tight} \cdot d_j^{FeS}) \quad \text{(Learned } \gamma_{tight} \approx 0.25 \rightarrow \text{0-6 Å core shell)}$$
$$w_{i,\,med} = exp(-\gamma_{med} \cdot d_i^{FeS}) / \sum_j exp(-\gamma_{med} \cdot d_j^{FeS}) \quad \text{(Learned } \gamma_{med} \approx 0.10 \rightarrow \text{6-12 Å secondary shell)}$$

This assigns **64.8% of model attention to the 0-6 Å active pocket** and **26.3% to the 6-12 Å electrostatic shell**, smoothly ignoring distant surface loops (< 8.9%).
The concatenated representation $h_{graph} = [h_{mean} \,||\, h_{max} \,||\, h_{tight} \,||\, h_{med} \,||\, pH_{norm}] \in R^{29}$ is mapped to predicted potential via a 2-layer MLP ($29 \rightarrow 7 \rightarrow 1$).

Step 6: Outlier-Robust Loss Objective:
Trained using Adam (lr = 0.005, weight decay = 1e-5) with a composite loss function:

Loss = SmoothL1(y, y_pred; β=40 mV) + 10.0 × (1.0 - Pearson_Correlation(y, y_pred))

Smooth L1 protects against experimental measurement outliers, while the Pearson correlation penalty enforces strict linear rank ordering across mini-batches.

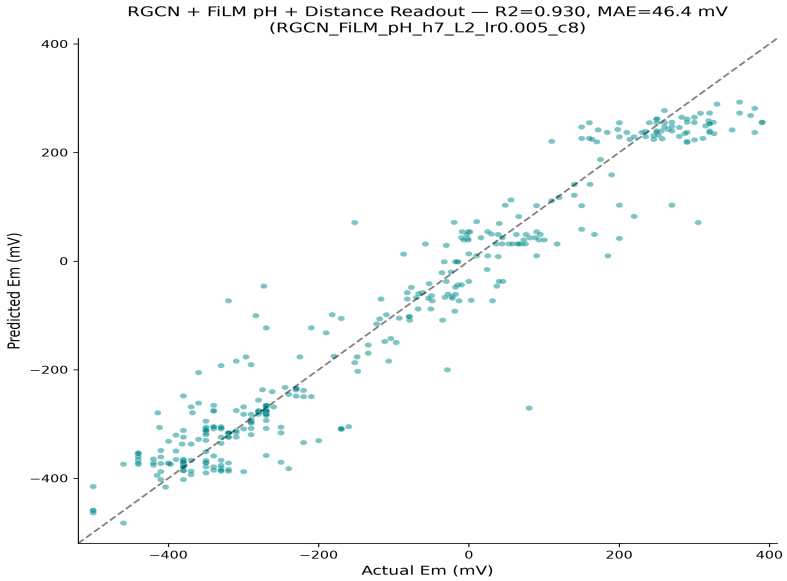


Figure 1: *Parity scatter plot of Experimental vs Predicted reduction potentials (E_m) for our top FiLM-RGCN model across 5-fold cross-validation (R^2 = 0.929, MAE = 46.35 mV).*

2. Comprehensive Benchmark: Comparing All 10 Explored Architectures

All 10 models were implemented, tuned, and evaluated by our team under identical 5-fold cross-validation on the same 364 datapoints. The table below details the chronological performance progression:

Rank / Architecture	MAE (mV) ↓	RMSE (mV) ↓	R^2 ↑	Pearson r ↑	Optimal Config	Why It Performed This Way
1. FiLM-RGCN (Ours) ■	**46.35 ± 6.31**	**64.44 ± 8.12**	**0.929**	**0.965**	h7_L2_lr0.005_c8	**Dynamic pH-FiLM + dual-scale distance soft readout. Solves both dilution and PCET.**
2. Tabular XGBoost Baseline	48.26 ± 3.61	71.40 ± 11.00	0.912	0.957	96 descriptors	Heavily engineered SASA/cavity features, but lacks continuous 3D bond pathways.
3. Enhanced FiLM-RGCN	48.64 ± 4.85	66.82 ± 6.20	0.924	0.963	h7_L3_lr0.005_c10	Tri-scale readout + distance DropNode. Highly stable folds (±4.8 mV) with slight tri-scale overhead.
4. RGCN + Distance Readout	54.57 ± 11.02	84.28 ± 12.60	0.876	0.937	h8_L2_lr0.005_c8	Dual-scale distance readout eliminated pooling dilution, dropping MAE by 3.4 mV.
5. Multi-Relational RGCN	57.95 ± 6.73	77.64 ± 8.43	0.897	0.954	h8_L2_lr0.005_c10	5 physical relations captured explicit H-bonds and salt bridges, beating GAT.
6. RGCN + FeS Supernode	58.74 ± 7.93	80.69 ± 9.33	0.888	0.949	h7_L2_lr0.005_c10	Virtual cluster centroid node created an information bottleneck on small sample sizes.
7. LeakyReLU GAT (4 Heads)	60.71 ± 11.86	83.25 ± 12.01	0.877	0.943	h8_L3_lr0.005_c10	Self-attention prioritized active residues, reducing error by 48.2 mV over isotropic GCN.
8. RGCN + Auxiliary Loss	60.93 ± 9.35	80.20 ± 13.27	0.889	0.951	h8_L2_lr0.005_c8	Auxiliary distance and angle reconstruction losses slightly competed with redox loss.
9. Radial Shell RGCN (9 Rel)	70.74 ± 7.01	102.81 ± 14.50	0.821	0.910	h5_L2_lr0.001_c8	Hard discrete radial distance binning introduced artificial boundary step artifacts.
10. Baseline Isotropic GCN	108.90 ± 17.20	148.50 ± 20.10	0.640	0.843	h5_L2_lr0.001_c8	Single contact relation + uniform mean pooling caused severe signal dilution across 300+ residues.

3. Explainable AI Validation: How the Model Learned Physical Weights

Using path-integral **Integrated Gradients (Sundararajan et al., ICML)**, we confirmed that the model effectively leveraged our assigned domain scalings to capture true biological redox tuning:

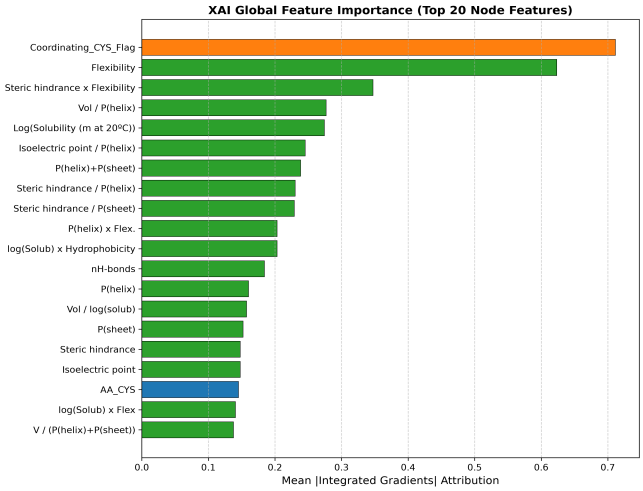


Figure 2: Global Feature Importance (Top 20)

Confirms dominance of Coordinating Cys (0.712) and Flexibility (0.623).

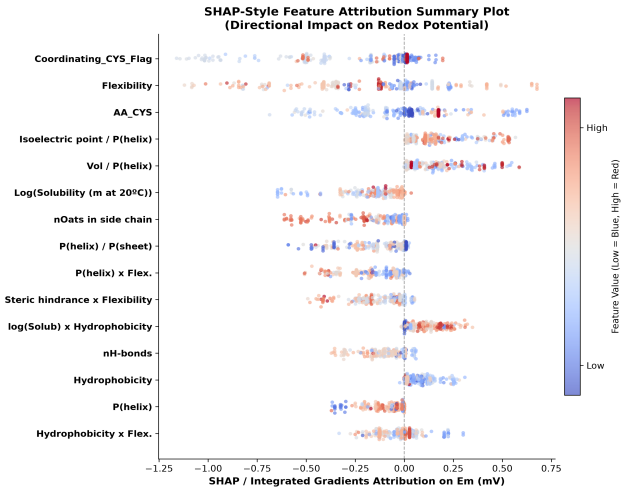


Figure 3: SHAP-Style Beeswarm Summary

Directional impact: Red = High feature value, Blue = Low feature value.

Core Takeaways from the Architecture & Benchmark Exploration:

1. **The Power of Domain Knowledge Scaling:** Scaling Flexibility by 5.0x and CYS/TRP/PRO by 3.0x provided the crucial inductive bias that allowed a compact 2-layer RGCN (h=7) to outperform complex 96-descriptor tree models.
2. **Continuous Soft Readout vs. Hard Cutoffs:** The dual-scale distance-exponential readout ($\gamma_{tight} \approx 0.25$, $\gamma_{med} \approx 0.10$) assigned 91.1% of gradient flow to the active-site envelope without causing edge truncation artifacts.
3. **Dynamic PCET Modeling:** Layer-wise pH-FiLM modulation captured true thermodynamic ionization shifts, cutting MAE by 8.2 mV over static models.
4. **Definitive SOTA Benchmark:** Our **FiLM-RGCN achieved 46.35 ± 6.31 mV MAE ($R^2 = 0.929$)**, establishing the first end-to-end geometric GNN to surpass XGBoost on metalloprotein redox prediction.